

DHRUV SINGH

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EDUCATION

Toronto Metropolitan University

Toronto, Canada

Bachelor of Engineering, Computer Engineering – Software Engineering Option (Co-op)

Sept 2023 – Apr 2028

- **3rd Place**, First-Year Engineering Student Design Competition – TMU (1 of 8 finalist teams selected from all first-year engineering sections), Nov 2023

EXPERIENCE

UI Technician

Toronto, Canada

Game Makers Union, TMU

Sept 2024 – Apr 2025

- Delivered **8 UI components** across 3 agile sprint cycles, collaborating with a 10+ member cross-functional team to maintain consistent design and development velocity.
- Analyzed user feedback from **25+ playtest sessions** and implemented iterative UI improvements, increasing task completion rate by **35%**.
- Profiled and resolved performance bottlenecks through systematic debugging, reducing frame drops by **40%** to ensure smooth, responsive gameplay.

Co-op Program Assistant

Toronto, Canada

Career, Co-op & Student Success Centre, TMU

Sept 2025 – Aug 2026

- Built **8 custom D2L (Brightspace) course shells** from scratch using HTML/CSS, designing and coding structured web pages as the digital front-end for co-op programs, deployed to **500+ students**.
- Migrated and provisioned **500+ student accounts** into the new shells, handling data mapping and access configuration for a seamless rollout with zero reported access issues.
- Managed and validated employment records for **200+ co-op students** using Excel (VLOOKUP, advanced formulas), maintaining **95%+ data integrity** across term reporting.
- Coordinated communications and check-in meetings across 5 program supervisors to keep student progress and shell rollout on track.

PROJECTS

Brainstormzz – AI-Powered Idea Structuring Platform

Dec 2025

- Developed a full-stack AI platform (React, Node.js, OpenAI API) converting unstructured input into hierarchical visual maps, onboarding **50+ beta users** within 2 weeks of launch.
- Conducted user research with **12 early-stage founders** to surface pain points in ideation workflows, translating insights into prioritized product requirements.
- Architected error handling and retry logic achieving a **98% successful generation rate**, reducing average time-to-insight by **60%** compared to traditional mind-mapping tools.

Carpool Matching Optimization System

Oct 2024

- Designed a route optimization algorithm matching **100+ drivers and riders** across 15 regional zones, reducing average commute distance by **22%**.
- Implemented multi-constraint satisfaction logic (non-smoking, gender preferences, detour limits, vehicle capacity) to ensure safety and user preference alignment across all test scenarios.
- Built a dual-metric evaluation framework measuring schedule alignment and CO₂ reduction, enabling data-driven matching decisions.

8-Bit General-Purpose Processor | VHDL, FSM, FPGA

Nov 2025 – Dec 2025

- Designed and implemented an 8-bit ALU and FSM-based control unit in VHDL, integrating register file, latches, and tri-state bus architecture with precise timing constraints.
- Validated the full design through simulation and FPGA deployment, achieving a **100%** match between simulated and hardware output across all instruction types.

Rover Arm Controller & CV Test Automation

Nov 2025 – Dec 2025

- Designed a socket-based Python control system for a 6-motor rover arm, translating live controller input into real-time motor command packets.
- Built a computer vision test automation framework validating ARUCO tag detection across **100+ test scenarios**, identifying and resolving **15+** defects through regression testing.

TECHNICAL SKILLS & COMPETENCIES

Web Development: HTML, CSS, JavaScript, React, Node.js, D2L/Brightspace (Course Shell Development), API Integration

Programming & Development: Python, Java, C, C#, SQL, VHDL, Git, GitHub

Tools: Excel, Jira, Figma, Quartus, ModelSim, OpenAI API, DBMS

Business & Analytical: Process Optimization, Data Validation & Quality Assurance, Stakeholder Coordination, Requirements Translation, Trade-off Analysis, Risk Assessment

Hardware & Systems: FPGA Design, Digital Circuits, FSM Implementation, Computer Vision (OpenCV), Real-time Control